10 - Implementation and Code Explanation

Module-by-module guide (parsed from the source)

Knowledge-pack rule: every statement below is derived from files in this project or is explicitly labelled as planned, missing or not verified. Nothing here is an invented citation, dataset, result or feature.

Source files used for this document

src/**/*.py; tests/*.py; analysis/, notebooks/, survey/ notebooks; infrastructure/scripts/*

How to read the code

Start at src/pipeline.py: run_pipeline() calls the stage packages in order. src/worker.py wraps it as a long-running service. Everything configurable lives in config/settings.yaml.

Modules

src/anomaly_detection/detectors.py (47 lines)

Unsupervised anomaly detection (no labels used for fitting).

Public name	Purpose (docstring first line)
fit_isolation_forest	(no docstring)
iso_scores	Higher = more anomalous.
fit_lof	(no docstring)
lof_scores	(no docstring)
evaluate_unsupervised	(no docstring)

src/common.py (64 lines)

Shared helpers: project paths, settings, structured (JSON) logging.

Public name	Purpose (docstring first line)
load_settings	(no docstring)
resolve	Absolute, created directory for a key under cfg['paths'].
JsonFormatter	One JSON object per line -> parsed by Promtail, queried in Loki.
setup_logging	(no docstring)
get_logger	(no docstring)

src/data_collection/external.py (23 lines)

Optional public benchmark: ULB/Worldline 'Credit Card Fraud Detection' dataset.

Public name	Purpose (docstring first line)
load_creditcard	(no docstring)

src/data_collection/synthetic.py (202 lines)

Isolated synthetic data generators (transactions, security events, survey).

Public name	Purpose (docstring first line)
generate_transactions	(no docstring)
generate_security_events	(no docstring)
pseudonymise	(no docstring)
generate_survey	SYNTHETIC respondents drawn from a latent-factor model (for pipeline demonstration).

src/database/db.py (68 lines)

PostgreSQL persistence (SQLite fallback for local runs and tests).

Public name	Purpose (docstring first line)
get_engine	(no docstring)
is_postgres	(no docstring)
init_schema	Apply database/schema.sql + migrations to PostgreSQL (idempotent: IF NOT EXISTS everywhere).
write_table	Append (or truncate+append) a DataFrame; failures are logged and counted, never fatal.
persist_run	(no docstring)

src/feature_engineering/features.py (85 lines)

Leakage-safe feature engineering for transactions and security events.

Public name	Purpose (docstring first line)
feature_names	(no docstring)
build_features	Return a numeric feature frame aligned with df's index.
chronological_split	(no docstring)
build_security_features	(no docstring)

src/fraud_detection/risk.py (26 lines)

Risk classification: map fraud probability to an operational action tier.

Public name	Purpose (docstring first line)
classify_risk	(no docstring)
tier_summary	(no docstring)

src/fraud_detection/train.py (85 lines)

Supervised fraud-detection models with randomised hyper-parameter search.

Public name	Purpose (docstring first line)
candidate_models	(no docstring)
train_and_select	Train every enabled model; tune on the training set (CV), pick threshold on validation.
score	(no docstring)
evaluate_models	(no docstring)

src/metrics/exporter.py (199 lines)

Prometheus instrumentation. Every value is computed by the pipeline - nothing hardcoded.

Public name	Purpose (docstring first line)
start_exporter	(no docstring)

Public name	Purpose (docstring first line)
instrument_stage	Time a pipeline stage and record success / failure. Yields a dict; set ['rows'] to export row counts.
publish_summary	Publish a run summary (dict produced by src.pipeline) to the Prometheus registry.

src/model_evaluation/evaluate.py (97 lines)

Binary-classification evaluation, plotting and threshold selection.

Public name	Purpose (docstring first line)
best_f1_threshold	(no docstring)
evaluate_binary	(no docstring)
bootstrap_ci	(no docstring)
comparison_table	(no docstring)
save_evaluation_plots	(no docstring)
save_feature_importance	(no docstring)

src/model_evaluation/statistics.py (75 lines)

Statistical analysis helpers (descriptive, hypothesis tests, correlations).

Public name	Purpose (docstring first line)
descriptive	(no docstring)
correlation	(no docstring)
correlation_with_p	(no docstring)
mann_whitney	(no docstring)
chi_square	(no docstring)
kruskal	(no docstring)
mcnemar	McNemar test (with continuity correction) on paired classifier errors.
ols	Ordinary least squares with classical standard errors on standardised predictors.

src/notebook_utils.py (60 lines)

Thin loaders so notebooks stay short and always read the pipeline's real artefacts.

Public name	Purpose (docstring first line)
load_raw	(no docstring)
load_clean	(no docstring)
load_features	(no docstring)
load_events	(no docstring)
load_survey	(no docstring)
load_trust	(no docstring)
load_summary	(no docstring)
load_model	(no docstring)
split_xy	(no docstring)

src/pipeline.py (410 lines)

End-to-end research pipeline.

Public name	Purpose (docstring first line)
save_df	(no docstring)
run_pipeline	(no docstring)
load_last_summary	(no docstring)
main	(no docstring)

src/preprocessing/anonymize.py (33 lines)

Privacy-conscious survey handling: pseudonymisation, generalisation, k-anonymity.

Public name	Purpose (docstring first line)
pseudonymise_ids	(no docstring)
drop_direct_identifiers	(no docstring)
k_anonymity	(no docstring)
enforce_k_anonymity	Suppress (drop) respondents belonging to quasi-identifier groups smaller than k.

src/preprocessing/cleaning.py (31 lines)

Cleaning / preprocessing of raw transaction and survey data.

Public name	Purpose (docstring first line)
clean_transactions	Return (clean_df, report). The report records every row removed and why.
frame_hash	Stable content hash used in model metadata for provenance.

src/preprocessing/validation.py (39 lines)

Data validation: returns structured check results (also exported as metrics).

Public name	Purpose (docstring first line)
validate_transactions	(no docstring)
validation_pass_rate	(no docstring)

src/reporting.py (83 lines)

Static research figures (PNG), an interactive Plotly comparison (HTML) and tables for the report.

Public name	Purpose (docstring first line)
make_figures	(no docstring)
write_tables	(no docstring)

src/schema.py (50 lines)

Canonical column names and categorical domains (single source of truth).

src/trust_scoring/scoring.py (70 lines)

Consumer Trust Score (CTS) and Platform Security Index (PSI).

Public name	Purpose (docstring first line)
reverse_code	(no docstring)
cronbach_alpha	(no docstring)
dimension_scores	(no docstring)
score_respondents	(no docstring)
reliability_report	(no docstring)
platform_security_index	(no docstring)

Public name	Purpose (docstring first line)
consumer_trust_index	(no docstring)

src/worker.py (111 lines)

Long-running service: runs the pipeline, exposes /metrics, then replays held-out data as a live stream.

Public name	Purpose (docstring first line)
main	(no docstring)

Key algorithms in plain language

Where	What happens
features.build_features	Per customer, running totals minus the current row give the average of PAST amounts; ratio of today's amount to that average flags unusual spending.
train.train_and_select	Randomly tries n_iter parameter sets, scores each by 3-fold average precision, refits the best, then picks the F1-optimal threshold on validation data.
evaluate.best_f1_threshold	Uses the precision-recall curve and takes the threshold with the highest F1.
detectors.fit_isolation_forest	Standardise, fit IsolationForest without labels; anomaly score = negative score_samples.
scoring.score_respondents	Reverse-code q12, rescale each dimension to 0-100, weighted average, segment by cut-offs.
exporter.publish_summary	Clears and refills labelled gauges from the run summary so Grafana always shows the latest run.
worker.main	Starts /metrics, runs the pipeline if artefacts are missing (or FORCE_RETRAIN=1), then loops: score a batch of held-out transactions, increment TP/FP/FN/TN, risk-tier and latency metrics.

Tests

Test file	Count	Test names
test_data_validation.py	5	test_dirty_synthetic_data_is_detected; test_missing_column_short_circuits; test_pass_rate_bounds; test_domain_violation_flagged; test_synthetic_generation_is_reproducible
test_fraud_detection.py	4	test_evaluate_binary_known_values; test_threshold_selection_beats_default_on_imbalance; test_risk_tiers; test_models_learn_better_than_chance_and_unsupervised_is_above_random
test_preprocessing.py	5	test_cleaning_removes_invalid_rows; test_features_have_stable_schema_and_no_nans; test_history_features_do_not_leak_future; test_chronological_split_is_ordered; test_survey_privacy
test_trust_scoring.py	9	test_score_bounds_and_reverse_coding; test_segments; test_weights_sum_to_one; test_cronbach_alpha_known; test_synthetic_survey_internal_consistency; test_indices; test_ols_recovers_coefficients; test_survey_driver_directions; test_hypothesis_helpers

Check	State
pytest result	**VERIFIED**

Notebooks

Notebook	Title	Code cells	Cells with output	Error outputs
analysis/consumer_trust_analysis.ipynb	Consumer trust analysis	7	7	0
analysis/cybersecurity_threat_analysis.ipynb	Cybersecurity threat analysis	7	7	0
analysis/data_quality_analysis.ipynb	Data quality analysis	6	6	0
analysis/exploratory_data_analysis.ipynb	Exploratory data analysis	7	7	0
analysis/fraud_pattern_analysis.ipynb	Fraud pattern analysis	5	5	0
analysis/statistical_analysis.ipynb	Statistical analysis	8	7	0
analysis/transaction_analysis.ipynb	Financial transaction analysis	8	7	0
notebooks/01_data_preprocessing.ipynb	01 - Data preprocessing	6	6	0
notebooks/02_feature_engineering.ipynb	02 - Feature engineering	6	6	0
notebooks/03_fraud_detection.ipynb	03 - Supervised fraud detection	6	6	0
notebooks/04_anomaly_detection.ipynb	04 - Unsupervised anomaly detection	6	5	0
notebooks/05_consumer_trust_scoring.ipynb	05 - Consumer Trust scoring	6	6	0
notebooks/06_model_training.ipynb	06 - Model training & hyper-parameter tuning	6	5	0
notebooks/07_model_evaluation.ipynb	07 - Model evaluation	8	7	0
notebooks/08_research_visualization.ipynb	08 - Research visualisation	7	7	0
survey/survey_analysis.ipynb	Survey analysis - consumer perceptions of FinTec	8	8	0

A notebook with zero cells containing outputs has been generated but not executed. Notebooks are generated by infrastructure/scripts/build_notebooks.py and executed by execute_notebooks.py.

Other scripts

Script	Purpose
infrastructure/scripts/generate_synthetic_data.py	generate datasets only
build_dashboards.py	regenerate Grafana JSON
build_notebooks.py	regenerate notebook sources
execute_notebooks.py	execute notebooks in place
initialize_database.sh	apply schema/migrations/seed outside Docker
run_pipeline.sh	run the pipeline

Incomplete or missing items

Item	Status
documentation/*.md written	IMPLEMENTED
Chapter 5 results and Chapter 6 conclusion in report/chapters	IMPLEMENTED
survey/survey_results.md	IMPLEMENTED
Grafana screenshots	MISSING
Automated test for exporter/worker	MISSING
CI configuration	MISSING